

Introduction Roughs for SPECTRUM

Drafted 2026-09-13; finalized after three rounds of adversarial review

This file contains three candidate Introductions for `01_slr_forecast_intervention2026.tex`, produced as directed: Version 1 follows the six-item bulleted outline already present at the top of that manuscript’s `\section{Introduction}`; Version 2 is an independently constructed introduction drawing on the broader survey in `literature_component_slr.tex` and the project’s own findings; Version 3 is a condensed synthesis of the two, retaining what we judged the most load-bearing points (the stakes, the three-way model-class trade-off, the WAIS tail-risk gap, and the SPECTRUM framework itself) while dropping supporting detail and secondary citations. All three are written to match the voice, quantitative style, and citation density of the target manuscript. Versions 1 and 2 were each drafted, then put through three rounds of independent adversarial review (fresh reviewers each round, checking citation validity against `references.bib`, accuracy against the papers actually cited, style conformance, and prose quality) with a full revision pass after each round, and were confirmed in the final round to compile cleanly (verified with a live `pdflatex/bibtex` run) with no outstanding material issues. Version 3 draws only on claims and citations already verified in Versions 1 and 2, and received one adversarial review pass of its own (citation validity, accuracy, and length target) rather than the full three rounds. Nothing here has been copied into the manuscript itself.

Length note. Version 1 is 1044 words and Version 2 is 872 words (body text only, excluding headings, counted programmatically). Version 3 originally targeted the intersection of “half to two-thirds” of each, i.e. $522\text{--}581$ words, and its first draft came in at 580 words, inside that band relative to both ($0.56\times V1$, $0.67\times V2$). A second revision pass (described below) added connective sentences to fix structural and clarity problems surfaced by adversarial review; the current text is 623 words ($0.60\times V1$, $0.71\times V2$), modestly above the original band, a trade-off made deliberately in favor of clarity over the length target. Its citation count (23) is unchanged by that pass and was scaled down roughly in proportion to its length rather than held to the 25–40 range specified for Versions 1 and 2, since forcing that many citations into a passage this short would come at the expense of prose quality. Version 3’s first draft received one full adversarial review pass, which caught (and that pass fixed) several compression artifacts: a fabricated causal link between the model-parsimony citations and the held-out-validation design, an inaccurate collapsing of Rahmstorf (2007) and Vermeer & Rahmstorf (2009) into a single-coefficient description (the latter has two), a dropped hierarchy-of-models rationale, an orphaned citation, and a reversed description of what component-level models trade away.

Revision note (second pass). The revised draft was then checked by three independent

adversarial reviewers, each rereading the full text and mapping every sentence to the idea it is meant to convey, one focused on argumentative structure and logical coherence, one on plain-language and sentence-level clarity, and one on whether each sentence’s content actually matched the goal that mapping assigned it. All three independently flagged the same unresolved referent (the phrase “that gap” opening Paragraph 3, which pointed to nothing named in the preceding paragraph); two flagged a component-level-models sentence that conflated a stated weakness (assuming stationarity) with an unrelated, more important claim (no published implementation withholds the aggregate record for an independent check) behind a single semicolon; and reviewers separately flagged an unearned closing claim in Paragraph 2 (that ordering model classes by complexity lets simpler ones check more complex ones, asserted without the paragraph showing how), an unsupported closing claim in Paragraph 3 (an appeal to “adaptation planning” introduced with no groundwork), and several sentences overloaded with two or more distinct claims joined by colons or semicolons. The text above fixes all of these: it names the referent explicitly (“this missing independent check”), splits the conflated and overloaded sentences, ties the Paragraph 2 closing claim to the specific held-out GMSLR–GMST check performed in Paragraph 4, and rewrites the Paragraph 3 closing claim to point back to the flooding-damage stakes already established in Paragraph 1 rather than to an unintroduced concept. No citations were added or removed in this pass.

Bibliography note. Version 1 cites only keys that already exist in `references.bib`. Version 2 additionally cites several papers discussed in `literature_component_slr.tex` that are not yet in `references.bib`: Felikson2025, Slangen2022, Garner2018, Nauels2017, Kopp2023, McCormack2026, Bakker2017, Darnell2025, and Held2005. These entries would need to be added to `references.bib` before Version 2 can be compiled or merged into the manuscript; per task instructions, no file other than this one has been modified.

Version 1 — follows the manuscript’s own itemized outline

Coastal communities plan against a moving target. Each additional centimeter of global mean sea-level rise (GMSLR) exposes roughly 1.6 million more people to annual flooding (*Kulp and Strauss*, 2019), nearly 90% of them in low- and middle-income countries (*Jongman et al.*, 2012). Without further adaptation, annual flood damages are projected to reach US\$14–18T (trillion) per year once GMSLR reaches 0.9–1.1 m (*Jevrejeva et al.*, 2018), while a lower-bound estimate of the cost of adapting to that rise, based on the price of raising dikes along a small fraction of the world’s coastlines, is on the order of US\$4T per meter (*Tiggeloven et al.*, 2020). Developing countries currently receive roughly US\$21B per year in adaptation finance across all sectors (*United Nations Environment Programme*, 2023), far short of what these cost estimates imply is needed. That shortfall makes the reliability of the sea-level projections used to plan adaptation directly consequential for how effectively it can be closed.

Sea-level projections have been published regularly for more than four decades and span a wide range of methods and outcomes (*Garner et al.*, 2026). Rates of GMSLR have already doubled twice since the mid-20th century, first over about 16 years and 0.34 °C of warming and again over about 21 years and 0.41 °C (*Frederikse et al.*, 2020; *Hamlington et al.*, 2024;

Rohde and Hausfather, 2020). Yet current coastal planning still leans heavily on projections from the IPCC Sixth Assessment Report (AR6), which places 21st-century rise at 0.4–0.7 m (*Fox-Kemper et al.*, 2021; *Hirschfeld et al.*, 2023). This variety is not a weakness so much as an underused resource: no single class of model is simultaneously the most physically detailed, the most transparent, and the most tightly constrained by data, and a framework that arrays these approaches by complexity, testing each rung against data before trusting the next, offers a way to combine their respective strengths and to identify where projections agree, diverge, or exceed what the data can support (*Green and Armstrong*, 2015; *Makridakis et al.*, 2020).

At the most detailed end sit numerical ice-sheet models, which resolve ice flow, deformation, and fracture directly and let competing physical hypotheses, such as whether marine ice-cliff instability can activate this century, be posed and debated within a common numerical framework (*DeConto and Pollard*, 2016; *Edwards et al.*, 2019; *Morlighem et al.*, 2024). Their principal weakness is that this physical detail demands more constraints than the observational record can currently supply. In hindcasting studies, process-based models diverge from historical GMSLR reconstructions (*Aschwanden et al.*, 2021) and from satellite observations within a decade of initialization (*Fricker et al.*, 2025). These disagreements trace to structural gaps such as the omission of leading-order calving physics (*Fricker et al.*, 2025) and an assumed ice viscosity far less sensitive to stress than observations indicate (*Millstein et al.*, 2022; *Getraer and Morlighem*, 2025), compounded by initialization uncertainty and, for marine ice sheets specifically, dynamics that can skew the resulting projections toward high outcomes rather than merely widening them (*Robel et al.*, 2019). The resulting parameterizations are numerous enough, and their interactions complex enough, that these models function largely as black boxes even to their developers, and every century-scale projection necessarily extrapolates them well beyond the short observational record used to tune them. Decades of general forecasting research show that models with more free parameters than their calibration data can constrain tend to forecast worse, not better, than simpler alternatives (*Green and Armstrong*, 2015).

At the opposite extreme are semi-empirical models that regress the observed rate of GMSLR directly onto global mean surface temperature (GMST), a class whose transparency and direct grounding in the data being predicted make it easy to test and to communicate (*Rahmstorf*, 2007; *Vermeer and Rahmstorf*, 2009). Statistically, semi-empirical fits track the observed sea-level record over multi-decade hindcast windows at least as well as, and at times better than, contemporaneous process-based projections (*Rahmstorf et al.*, 2012), and Bayesian treatments of the same GMST-GMSL relationship, which propagate measurement and proxy uncertainty explicitly rather than fitting a single point estimate, extend the approach over far longer records (*Kopp et al.*, 2016). Their central limitation is that a single global regression coefficient is not obviously physical or causal, may not stay stationary as individual components respond to different forcings on different timescales, and gives no way to tell, when it forecasts correctly, whether that agreement reflects the right physical reasons (*Church et al.*, 2013).

Between these extremes sit component-level models that decompose GMSLR into its physical contributors, ocean thermal expansion, glaciers, and the ice sheets, and calibrate each against existing observational or model data (*Kopp et al.*, 2014; *Mengel et al.*, 2016; *Wong et al.*, 2017). This intermediate complexity is a deliberate trade. Parsimony and a

direct connection to physical processes come at the cost of lumped parameters that cannot resolve individual basins or glaciers, an assumption of stationarity over the calibration period, and the same extrapolation risk that limits every other class of model. The calibration strategy within this class still varies: some scale a component decomposition to modeled temperature trajectories (*Perrette et al.*, 2013), others use Bayesian-calibrated reduced-complexity models, with an expert prior on the most uncertain term (Antarctic dynamics), to bound a physically plausible upper limit on total sea-level rise (*Sriver et al.*, 2012), and still others forgo new component physics altogether and instead statistically fuse existing probabilistic projections into a single consensus estimate (*Grandey et al.*, 2024).

A framework that spans this hierarchy, from semi-empirical regression through component-level, data-calibrated models to full process-based simulation, does not need to choose a single class as authoritative. It can use agreement or disagreement across rungs as a diagnostic in its own right, letting the simplest, most transparent models motivate and check the more complex ones, calibrating and validating within the space of models rather than against a single preferred one, and it can let the most consequential insights come from the simplest rung rather than the most complex. Here, we construct such a framework: a data-driven, component-level sea-level projection that treats each contributor to GMSLR, ocean thermal expansion, Greenland Ice Sheet (GrIS) surface mass balance (SMB) and glacier discharge, West Antarctic Ice Sheet (WAIS) glacier discharge, East Antarctic Ice Sheet (EAIS) SMB, the Antarctic Peninsula, glaciers outside the ice sheets, and terrestrial water storage, with its own parsimonious physical model, calibrated against independent observational records using Bayesian inference. We call this framework SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. Most of its component models adapt existing formulations from the literature; the exception is Greenland Ice Sheet discharge, whose 3-parameter model connecting glacier discharge directly to ocean temperature is developed here. Critically, observed GMSLR and its sensitivity to observed GMST are withheld from every component-level calibration and used only as diagnostics on the composite model, so that the framework’s central test, whether independently calibrated parts sum to reproduce a whole they were never fit to, is genuinely out of sample.

Version 2 — independent synthesis from the literature review

Every additional centimeter of global mean sea-level rise (GMSLR) exposes roughly 1.6 million more people to annual coastal flooding (*Kulp and Strauss*, 2019), and unmitigated damages are projected to reach US\$14–18T (trillion) per year once rise reaches 0.9–1.1 m (*Jevrejeva et al.*, 2018). Sea-level projections meant to guide adaptation against those costs have been published for more than four decades. A database of 756 such projections drawn from 103 studies shows their production concentrated in a handful of wealthy countries and their horizons only recently extended past 2100, and finds that, prior to the IPCC’s Sixth Assessment Report (AR6), individual studies’ high-end estimates routinely exceeded the IPCC’s own central ranges (*Garner et al.*, 2026). Over that same period, successive IPCC assessment cycles have revised their component treatments and uncertainty structure substantially (*Slangen et al.*, 2023; *Garner et al.*, 2018), evidence less of an immature field than

of one still searching for the right relationship between physical detail and the observational record available to constrain it.

A growing, cross-institutional view holds that the way forward is not more physical detail but tighter observational constraint. A recent synthesis spanning the leading process-based and observational sea-level research groups argues explicitly for narrowing projections with data and for characterizing uncertainty structure, not just magnitude, in both the observations and the models (*Felikson et al.*, 2025). That argument responds to a specific, well-documented weakness of the process-based ice-sheet models that dominate current assessments, including AR6’s own 0.4–0.7 m projection for 21st-century rise (*Fox-Kemper et al.*, 2021). In hindcasting exercises, these models diverge from the historical GMSLR record (*Aschwanden et al.*, 2021) and from satellite observations within a decade of initialization (*Fricker et al.*, 2025), disagreements traced to structural gaps such as an assumed ice viscosity far less sensitive to stress than observations indicate (*Millstein et al.*, 2022; *Getraer and Morlighem*, 2025).

Component-summation frameworks, which decompose GMSLR into its physical contributors and combine them probabilistically, already offer a partial answer, and existing implementations differ mainly in what each component is calibrated against: expert-assessed probability distributions (*Kopp et al.*, 2014), independently calibrated semi-empirical response functions (*Mengel et al.*, 2016), jointly calibrated physical models (*Wong et al.*, 2017), or reduced-complexity emulators of process-model output (*Nauels et al.*, 2017), with the modular platform now underlying AR6 designed to accommodate all of these calibration strategies within a common structure (*Kopp et al.*, 2023). What none of them do is hold out the aggregate observational record, global sea level’s own sensitivity to global temperature, entirely from every component-level fit and use it solely as an independent check on the summed result.

The stakes of that calibration choice are concentrated in West Antarctica. Marine ice sheet instability does not simply widen the uncertainty on Antarctic contributions; it mathematically skews the resulting distribution toward high outcomes (*Robel et al.*, 2019), and independent regional ocean modeling finds that ocean-driven melt of West Antarctic ice shelves accelerates at similar rates across emissions scenarios this century, meaning much of that risk may already be locked in (*Naughten et al.*, 2023). Recent work also finds that the near-term rate of Antarctic mass loss is itself predictive of the rate over the following 3–5 decades (*McCormack et al.*, 2026), suggesting the tail is not only large but partially foreseeable from the current observational record. Yet deep-uncertainty treatments that represent this risk as a structured scenario mixture rather than a single process-model trajectory remain rare (e.g., *Bakker et al.*, 2017). A recent variance decomposition reaches a seemingly contrary conclusion, that emissions-scenario choice dominates projection uncertainty through the mid-century before geophysical uncertainty takes over (*Darnell et al.*, 2025), a claim that a framework treating the tail explicitly cannot simply set aside.

None of this is an argument against physical detail; it is an argument for testing that detail against data before trusting it. Outside sea-level rise specifically, climate science has long recognized that a hierarchy spanning idealized and comprehensive models serves understanding, and often forecast robustness, better than comprehensive models alone (*Held*, 2005), a pattern general forecasting research confirms empirically across some 100,000 time series (*Green and Armstrong*, 2015; *Makridakis et al.*, 2020). Semi-empirical models that regress

GMSLR directly onto GMST provide another genuinely data-driven benchmark (*Rahmstorf, 2007; Vermeer and Rahmstorf, 2009*), distinguishing the choice of model complexity from the choice of calibration source. Nor is the record uniformly unkind to assessment-based projection. An earlier IPCC assessment’s mid-range 21st-century sea-level projection verified to within about a centimeter three decades later (*Törnqvist et al., 2025*), a reminder that any critique of process-based projections should be aimed at specific assessment cycles and specific components, not at process-based reasoning as such.

Here, we build a component-level framework that puts that discipline into practice directly: every physical contributor to GMSLR, ocean thermal expansion, Greenland Ice Sheet (GrIS) surface mass balance and glacier discharge, West Antarctic Ice Sheet (WAIS) glacier discharge, East Antarctic Ice Sheet (EAIS) surface mass balance, the Antarctic Peninsula, glaciers outside the ice sheets, and terrestrial water storage, is calibrated with its own parsimonious physical model against an independent observational record, using Bayesian inference to propagate parameter uncertainty rather than adopting it from assessment or process-model output. We call the result SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. The only genuinely new component model links GrIS discharge to ocean temperature through a 3-parameter time-lagged response; every other component adapts an existing formulation. Its treatment of WAIS is a structured deep-uncertainty scenario mixture built to represent the skewed tail described above rather than average over it. Observed GMSLR and its GMST sensitivity, withheld from every component calibration, are reserved as the composite-level diagnostic that current frameworks do not report, and against which SPECTRUM’s aggregated components are tested here for the first time.

Version 3 — condensed synthesis

Global mean sea-level rise (GMSLR) already exposes roughly 1.6 million more people to annual flooding for every additional centimeter (*Kulp and Strauss, 2019*), with unmitigated damages projected at US\$14–18T (trillion) per year once rise reaches 0.9–1.1 m (*Jevrejeva et al., 2018*). Sea-level projections meant to inform that risk have been published for four decades and span a wide range of methods and outcomes (*Garner et al., 2026*). Rates of GMSLR have already doubled twice since the mid-20th century (*Frederikse et al., 2020; Hamlington et al., 2024*), yet coastal planning still leans heavily on the IPCC Sixth Assessment Report (AR6), which projects 0.4–0.7 m of 21st-century rise (*Fox-Kemper et al., 2021; Hirschfeld et al., 2023*). A growing, cross-institutional view holds that narrowing AR6’s range requires tighter observational constraint on projections, not more physical detail (*Felikson et al., 2025*).

Three model classes have taken up that challenge, each with a real weakness. Process-based ice-sheet models resolve ice flow and fracture directly, but they demand more constraints than the observational record can currently supply. In hindcasting studies, they diverge from the historical GMSLR record (*Aschwanden et al., 2021*) and from satellite observations within a decade of initialization (*Fricker et al., 2025*). These disagreements trace to gaps such as an assumed ice viscosity far less sensitive to stress than observations indicate (*Millstein et al., 2022; Getraer and Morlighem, 2025*). Decades of general forecasting research

show that models with more free parameters than their data can constrain tend to forecast worse than simpler alternatives (*Green and Armstrong, 2015; Makridakis et al., 2020*). Semi-empirical models that regress the rate of GMSLR onto global mean surface temperature are simple and transparent, but the underlying relationship is not obviously physical, causal, or stationary (*Rahmstorf, 2007; Vermeer and Rahmstorf, 2009; Church et al., 2013*). Even so, such fits track the observed record at least as well as contemporaneous process-based projections (*Rahmstorf et al., 2012*). Between these extremes, component-level models decompose GMSLR into its physical contributors and calibrate each against existing data (*Kopp et al., 2014; Mengel et al., 2016; Wong et al., 2017*), trading basin-scale resolution for parsimony and a direct physical connection, at the cost of assuming stationarity. None of the published implementations, however, withholds the aggregate observational record from every component fit, so none can use it as an independent check on the summed result. Ordering these three model classes by complexity, rather than treating any one as authoritative, turns their disagreement into a diagnostic, letting the simplest, most transparent models motivate and check the more complex ones.

The stakes of this missing independent check are concentrated in West Antarctica, where marine ice sheet instability does not merely widen projection uncertainty but skews it toward high outcomes (*Robel et al., 2019*), and where independent regional ocean modeling finds melt accelerating at similar rates across emissions scenarios this century, meaning much of that risk may already be locked in (*Naughten et al., 2023*). This West Antarctic risk differs structurally from the temperature-sensitive uncertainty in the other components. Treating it as a single central trajectory, rather than a mixture spanning its full range, would understate exactly the high-end damages introduced above.

Here, we build a component-level framework in which each contributor to GMSLR (ocean thermal expansion; Greenland Ice Sheet (GrIS) surface mass balance and glacier discharge; West Antarctic Ice Sheet (WAIS) glacier discharge; East Antarctic Ice Sheet (EAIS) surface mass balance; the Antarctic Peninsula; glaciers outside the ice sheets; and terrestrial water storage) gets its own parsimonious physical model, calibrated against an independent observational record using Bayesian inference. We call it SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. Its only new component model links GrIS discharge to ocean temperature through a 3-parameter time-lagged response. Its treatment of WAIS is a structured deep-uncertainty scenario mixture, built to represent the skewed tail described above rather than average over it. Observed GMSLR and its sensitivity to GMST, the relationship at the core of the semi-empirical class described above, are withheld from every component calibration and used only as a diagnostic on the composite model. The central test is therefore out of sample. It asks whether the independently calibrated parts reproduce a whole they were never fit to.

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